

INTERNATIONAL GCSE

Biology

9201/1

Paper 1

Mark Scheme

November 2021

Version: 1.0 Final Mark Scheme



Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Assessment Writer.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

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Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Question	Answers	Extra information	Mark
1.1	tongue eye ear	allow mouth allow retina allow cochlea	1 (1-3) 1 (1-3) 1 (1-3)

Question	Answers	Extra information	Mark
1.2	stimulus coordinator effector response	in this order only all correct = 3 marks 2 or 3 correct = 2 marks 1 correct = 1 mark	3 (1-3)

Question	Answers	Extra information	Mark
1.3	as a chemical which moves across the gap / synapse / junction and causes an impulse (in the next neurone) or which diffuses across the gap / synapse (to the next neurone)	allow neurotransmitter / acetylcholine	1 (4-5) 1 (6-7)

		Total	8
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Question	Answers	Extra information	Mark
2.1	Flattened body Hard outer case Hooks on ends of legs Sharp mouth parts Can bite to feed on cat blood Can grip fur so it holds on firmly Can move easily through the fur Cannot be easily damaged all correct = 3 marks, 2 or 3 correct = 2 marks, 1 correct = 1 mark do not award mark if additional line drawn from one adaptation		3 (1-3)

Question	Answers	Extra information	Mark
2.2	structural		1 (1-3)

Question	Answers	Extra information	Mark
2.3	parasite host	in this order only	1 (1-3) 1 (4-5)

Question	Answers	Extra information	Mark
2.4	amino acid(s)		1 (6-7)

Question	Answers	Extra information	Mark
2.5	extremophile(s)		1 (4-5)
		Total	8

Question	Answers	Extra information	Mark
3.1	plasma		1 (1-3)

Question	Answers	Extra information	Mark
3.2	white blood cells		1 (1-3)

Question	Answers	Extra information	Mark
3.3	urea		1 (1-3)

Question	Answers	Extra information	Mark
3.4	carbon dioxide		1 (1-3)

Question	Answers	Extra information	Mark
3.5	to transport oxygen to the organs		1 (1-3)

Question	Answers	Extra information	Mark
3.6	platelets / small fragments of cells (stick to the wound site)		1 (4-5)
	starting a series of enzyme- controlled reactions	allow which causes prothrombin to convert to thrombin	1 (6-7)
	which results in fibrinogen changing to fibrin		1 (6-7)
	forming a network of fibres		1 (6-7)
	which trap the blood cells and form the clot / scab		1 (6-7)
		Total	10

Question	Answers	Extra information	Mark
4.1	heart rate		1 (4-5)

Question	Answers	Extra information	Mark
4.2	any two from: - age - sex - exercise rate - rest time between each exercise	allow gender allow description of exercising at same rate eg. 1 step every 10 secs allow rest until heart rate returns to normal	2 (4-5)

Question	Answers	Extra information	Mark
4.3	method for counting heart rate / pulse method to measure time or record beats per unit time		1 (4-5) 1 (4-5)

Question	Answers	Extra information	Mark
4.4	x- axis labelled number of step ups and y- axis labelled heart rate in beats per minute suitable scale for y - axis all points correct suitable smooth curve of best fit	allow tolerance + / - half a small square allow 1 mark for 3 or 4 points correct	1 (4-5) 1 (4-5) 2 (6-7) 1 (6-7)

Question	Answers	Extra information	Mark
4.5	as the level of exercise increases so does the heart rate		1 (4-5)
	until it reaches a maximum at 15 step-ups		1 (6-7)
	or until it reaches a maximum at 174 beats per minute		

Question	Answers	Extra information	Mark
4.6	fatty material / fat has built up inside the artery		1 (6-7)
	restricting the flow of blood	allow blocking the blood flow	1 (6-7)
	so not enough oxygen / glucose reaches the heart muscle / cells		1 (6-7)

Question	Answers	Extra information	Mark
4.7	a stent		1 (4-5)

		Total	16
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Question	Answers	Extra information	Mark
5.1	$3.1 - 2.5 = 0.6$ $(0.6 \div 2.5) \times 100$ $= 24 (\%)$		1 (4-5) 1 (4-5) 1 (4-5)

Question	Answers	Extra information			Mark	
5.2			Characteristic			2 (1-3) 2 (4-5)
			Male sex	Brown eyes	Birth weight	
	Cause of characteristic	Genetic only	✓	✓		
		Environmental only				
		Genetic and environmental			✓	
	Type of variation	Continuous			✓	
		Discontinuous	✓	✓		
	1 mark for each correct tick.					
	do not award mark if more than one tick in each section for each characteristic.					

Question	Answers	Extra information	Mark
5.3	father has gene(s) / allele(s) for brown eyes (which) passed into his sperm / gamete cell which fertilised egg (then) mitosis occurred so each embryo has same genes / alleles	ignore references to brown eyes being dominant over blue eyes.	1 (4-5) 1 (6-7) 1 (6-7) 1 (8-9)

Question	Answers	Extra information	Mark
5.4	differentiation		1 (6-7)

Question	Answers	Extra information	Mark
5.5	(advantage -) will not be attacked by the patient's antibodies or patient will not need immunosuppressant drugs (because) cells will have same (tissue) antigens on the surface (disadvantage -) involves a (human) embryo which raises ethical concerns	allow will not be rejected by (the patient's) immune system allow cells will be a tissue match / type if no other mark awarded for disadvantage allow example e.g. embryo killed	1 (6-7) 1 (8-9) 1 (4-5) 1 (4-5)

Question	Answers	Extra information	Mark
5.6	bone marrow	allow other examples eg blood, skin	1 (6-7)

Question	Answers	Extra information	Mark
5.7	cells placed in patient's body / pancreas (where they) produce insulin to control the blood glucose concentrations or to move glucose from the blood into the cells or to convert glucose to glycogen		1 (6-7) 1 (4-5) 1 (4-5)
		Total	20

Question	Answers	Extra information	Mark
6.1	Transport process Active transport Osmosis do not award mark if additional line drawn from one process	Definition The movement of water from a concentrated solution to a dilute solution through a partially permeable membrane The movement of water from a dilute solution to a concentrated solution through a partially permeable membrane The movement of particles from a low concentration to a high concentration requiring energy The spreading out of particles when concentrations are equal	1 (4-5) 1 (6-7)

Question	Answers	Extra information	Mark
6.2	diffusion		1 (4-5)

Question	Answers	Extra information	Mark
6.3	any four from - villi increase the surface area (for greater absorption / diffusion / transport of molecules) - microvilli further increase the surface area (for greater absorption / diffusion / transport of molecules) (layer of cells covering villi are) only one cell thick which provides a short diffusion path - a large capillary network (takes away absorbed molecules) which increases	allow capillary walls are only one cell thick which provides a short diffusion path allow the capillary network provides oxygen and glucose to release energy by respiration for	2 (6-7) 2 (8-9)

	the concentration gradient for diffusion (into the blood) cells covering villi have (many) mitochondria to transfer / release energy for active transport	active transport allow other correct transport processes that require energy	
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Question	Answers	Extra information	Mark
6.4	Level 3: Relevant points are identified, given in detail and logically linked to form a clear account.		5–6 (6-7)
	Level 2: Relevant points are identified, and there are attempts at linking. The resulting account is not fully clear.		3–4 (6-7)
	Level 1: Relevant points are identified and stated simply, but their relevance is not clear and there is no attempt at logical linking.		1–2 (4-5)
	No relevant content		0
	Indicative content Digestion - protein is broken down by protease - protein is broken down to amino acid(s) - this is shown by the (large) increase in amino acids in the small intestine after the meal - carbohydrate is broken down by amylase / carbohydrase - carbohydrate is broken down to glucose / sugar molecules - this is shown by the (large) increase in glucose in the small intestine after the meal - the small intestine contains protease, amylase / carbohydrase (and water) so the digestion can take place here (but) there is amylase in the saliva / mouth and protease in the stomach so some was probably digested before it reached the small intestine Transport - starch / carbohydrate molecules are large / insoluble (so) can't pass through the membrane - this is shown by there being zero starch in the blood - amino acids are absorbed against the concentration gradient when fasting (570 compared to 1250) so it must be active transport		

	• after the meal the concentration of amino acids is greater in the intestine than the blood (9500 compared to 1500) so it is transported by diffusion • same arguments for glucose as amino acids quoting concentrations. For level 3 students must refer to digestion of proteins and carbohydrate, refer to transport and give comparative data.	
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Question	Answers	Extra information	Mark
6.5	(due to the increased concentration of her blood) the pituitary gland released ADH		1 (8-9)
	which caused increased reabsorption of water in the kidney (tubules)		1 (8-9)

		Total	15
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Question	Answers	Extra information	Mark
7.1 view with Figure 7	eating / feeding		1 (4-5)

Question	Answers	Extra information	Mark
7.2 mark with 7.3	$6\text{CO}_2 + 6\text{H}_2\text{O}$ $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$	allow 1 mark for correct formulae but not balanced	1 (6-7) 1 (6-7)

Question	Answers	Extra information	Mark
7.3 mark with 7.2	$\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$ $6\text{CO}_2 + 6\text{H}_2\text{O}$	allow 1 mark for correct formulae but not balanced allow 2 marks if equations are correct but interchanged between 7.2 and 7.3	1 (6-7) 1 (6-7)

Question	Answers	Extra information	Mark
7.4	correct graph readings 10.2 and 6.0 4.2 (a.u.)	allow $\pm$ half small square	1 (4-5) 1 (4-5)

Question	Answers	Extra information	Mark
7.5	two sets of correct readings from the line	for example - 34.5, 11.8 and 40, 10.2	1 (8-9)
	$\frac{\Delta y}{\Delta x}$ for the line	for example $\frac{1.6}{(-) 5.5}$	1 (8-9)
	gradient	(-) 0.29 (a.u. / °C) allow values between (-) 0.28 and (-) 0.30	1 (8-9)

Question	Answers	Extra information	Mark
7.6	activity of enzymes responsible for decay increases with temperature	allow increased temperature causes increased molecular movement leading to faster reaction rate of enzymes	1 (8-9)
	or enzyme reactions in microorganisms increase with temperature up to 38°C after which rate falls due to denaturation (of enzymes)	allow description of denaturation	1 (8-9)

Question	Answers	Extra information	Mark
7.7	(they need to be added and removed by processes) at equal rates	allow processes / rates need to be in equilibrium or processes / rates need to be balanced	1 (6-7)

		Total	13
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